

STA304 — Final Exam Formula Sheet

Simple Random Sampling (SRS)

	Estimator	Estimated Variance	Confidence Interval
Mean	$\bar{y}_S = \frac{1}{n} \sum_{i \in S} y_i$	$\hat{V}(\bar{y}_S) = \frac{s_y^2}{n} \left(1 - \frac{n}{N}\right)$	$\bar{y}_S \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\bar{y}_S)}$
Total	$\hat{t}_S = N \bar{y}_S$	$\hat{V}(\hat{t}_S) = N^2 \frac{s_y^2}{n} \left(1 - \frac{n}{N}\right)$	$\hat{t}_S \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\hat{t}_S)}$
Proportion	$\hat{p}_S = \frac{1}{n} \sum_{i \in S} y_i$	$\hat{V}(\hat{p}_S) = \frac{\hat{p}_S(1 - \hat{p}_S)}{n - 1} \left(1 - \frac{n}{N}\right)$	$\hat{p}_S \pm z_{\alpha/2} \sqrt{\hat{V}(\hat{p}_S)}$

y_i are binary (0/1) for proportion s_y^2 = sample variance of y

Stratified Random Sampling (STRS)

	Estimator	Estimated Variance	Confidence Interval
Mean	$\bar{y}_{\text{str}} = \sum_{h=1}^H \frac{N_h}{N} \bar{y}_{hS}$	$\hat{V}(\bar{y}_{\text{str}}) = \sum_{h=1}^H \left(1 - \frac{n_h}{N_h}\right) \left(\frac{N_h}{N}\right)^2 \frac{s_{yh}^2}{n_h}$	$\bar{y}_{\text{str}} \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\bar{y}_{\text{str}})}$
Total	$\hat{t}_{y, \text{str}} = \sum_{h=1}^H N_h \bar{y}_{hS}$	$\hat{V}(\hat{t}_{y, \text{str}}) = \sum_{h=1}^H \left(1 - \frac{n_h}{N_h}\right) N_h^2 \frac{s_{yh}^2}{n_h}$	$\hat{t}_{y, \text{str}} \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\hat{t}_{y, \text{str}})}$
Proportion	$\hat{p}_{\text{str}} = \sum_{h=1}^H \frac{N_h}{N} \hat{p}_{hS}$	$\hat{V}(\hat{p}_{\text{str}}) = \sum_{h=1}^H \left(1 - \frac{n_h}{N_h}\right) \left(\frac{N_h}{N}\right)^2 \frac{\hat{p}_{hS}(1 - \hat{p}_{hS})}{n_h - 1}$	$\hat{p}_{\text{str}} \pm z_{\alpha/2} \sqrt{\hat{V}(\hat{p}_{\text{str}})}$

$\bar{y}_{hS}$ = sample mean within stratum h s_{yh}^2 = sample variance of y within stratum h $\hat{p}_{hS}$ = sample proportion within stratum h

Ratio Estimation (SRS)

	Estimator	Estimated Variance	Confidence Interval
Ratio	$\hat{B} = \frac{\bar{y}_S}{\bar{x}_S}$	$\hat{V}(\hat{B}) = \left(1 - \frac{n}{N}\right) \frac{s_e^2}{n\bar{x}_S^2}$	$\hat{B} \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\hat{B})}$
Mean	$\hat{y}_r = \hat{B} \bar{x}_U$	$\hat{V}(\hat{y}_r) = \left(1 - \frac{n}{N}\right) \left(\frac{\bar{x}_U}{\bar{x}_S}\right)^2 \frac{s_e^2}{n}$	$\hat{y}_r \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\hat{y}_r)}$
Total	$\hat{t}_{yr} = \hat{B} t_x$	$\hat{V}(\hat{t}_{yr}) = \left(1 - \frac{n}{N}\right) \left(\frac{t_x}{\bar{x}_S}\right)^2 \frac{s_e^2}{n}$	$\hat{t}_{yr} \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\hat{t}_{yr})}$

$$s_e^2 = \frac{1}{n-1} \sum_{i \in S} (y_i - \hat{B} x_i)^2 \quad \bar{x}_U = \text{known population mean of } x \quad t_x = N\bar{x}_U = \text{known population total of } x$$

Ratio Estimation with Stratified Sampling

	Estimator	Estimated Variance	Confidence Interval
Mean	$\hat{y}_{rc} = \hat{B} \bar{x}_U$	$\hat{V}(\hat{y}_{rc}) = \frac{\bar{x}_U^2}{\bar{x}_{str}^2} \sum_{h=1}^H \frac{N_h^2}{N^2} \left(1 - \frac{n_h}{N_h}\right) \frac{s_{eh}^2}{n_h}$	$\hat{y}_{rc} \pm z_{\alpha/2} \sqrt{\hat{V}(\hat{y}_{rc})}$
Total	$\hat{t}_{yrc} = \hat{B} t_x$	$\hat{V}(\hat{t}_{yrc}) = \frac{t_x^2}{\hat{t}_{x, str}^2} \sum_{h=1}^H N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_{eh}^2}{n_h}$	$\hat{t}_{yrc} \pm z_{\alpha/2} \sqrt{\hat{V}(\hat{t}_{yrc})}$

$$\hat{B} = \frac{\hat{t}_{y, str}}{\hat{t}_{x, str}} = \frac{\bar{y}_{str}}{\bar{x}_{str}} \quad s_{eh}^2 = s_{yh}^2 - 2\hat{B}s_{xyh} + \hat{B}^2 s_{xh}^2 \quad s_{xyh} = \text{sample covariance of } x \text{ and } y \text{ in stratum } h$$

Regression Estimation

	Estimator	Estimated Variance	Confidence Interval
Mean	$\hat{y}_{reg} = \hat{B}_0 + \hat{B}_1 \bar{x}_U$	$\hat{V}(\hat{y}_{reg}) = \left(1 - \frac{n}{N}\right) \frac{s_e^2}{n}$	$\hat{y}_{reg} \pm t_{\alpha/2, n-2} \sqrt{\hat{V}(\hat{y}_{reg})}$
Total	$\hat{t}_{yreg} = N\hat{B}_0 + \hat{B}_1 t_x$	$\hat{V}(\hat{t}_{yreg}) = \left(1 - \frac{n}{N}\right) \frac{N^2 s_e^2}{n}$	$\hat{t}_{yreg} \pm t_{\alpha/2, n-2} \sqrt{\hat{V}(\hat{t}_{yreg})}$

$$\hat{B}_0 = \bar{y}_S - \hat{B}_1 \bar{x}_S \quad s_e^2 = \frac{1}{n-1} \sum_{i \in S} (y_i - \hat{B}_0 - \hat{B}_1 x_i)^2 \quad \bar{x}_U = \text{known population mean of } x \quad t_x = N\bar{x}_U = \text{known population total of } x$$

Estimation in Domain

	Estimator	Estimated Variance	Confidence Interval
Mean	$\bar{y}_d = \frac{1}{n_d} \sum_{i \in S_d} y_i$	$\hat{V}(\bar{y}_d) = \left(1 - \frac{n}{N}\right) \frac{n}{n_d^2} \frac{(n_d - 1) s_{yd}^2}{n - 1}$	$\bar{y}_d \pm t_{\alpha/2, n_d-1} \sqrt{\hat{V}(\bar{y}_d)}$
Total	$\hat{t}_{yd} = N_d \bar{y}_d$	$\hat{V}(\hat{t}_{yd}) = \left(1 - \frac{n}{N}\right) \frac{n N_d^2}{n_d^2} \frac{(n_d - 1) s_{yd}^2}{n - 1}$	$\hat{t}_{yd} \pm t_{\alpha/2, n_d-1} \sqrt{\hat{V}(\hat{t}_{yd})}$

$s_{yd}^2 = \frac{1}{n_d - 1} \sum_{i \in S_d} (y_i - \bar{y}_d)^2$ $S_d = \text{sample units in domain } d$ $n_d = \text{sample size in domain } d$ $N_d = \text{domain population size}$

Post-Stratification

	Estimator	Estimated Variance	Confidence Interval
Mean	$\bar{y}_{\text{post}} = \sum_{h=1}^H \frac{N_h}{N} \bar{y}_{hS}$	$\hat{V}(\bar{y}_{\text{post}}) = \left(1 - \frac{n}{N}\right) \sum_{h=1}^H \frac{N_h}{N} \frac{s_{yh}^2}{n}$	$\bar{y}_{\text{post}} \pm z_{\alpha/2} \sqrt{\hat{V}(\bar{y}_{\text{post}})}$
Total	$\hat{t}_{y\text{post}} = \sum_{h=1}^H N_h \bar{y}_{hS}$	$\hat{V}(\hat{t}_{y\text{post}}) = \left(1 - \frac{n}{N}\right) \sum_{h=1}^H \frac{N_h N s_{yh}^2}{n}$	$\hat{t}_{y\text{post}} \pm z_{\alpha/2} \sqrt{\hat{V}(\hat{t}_{y\text{post}})}$

$\bar{y}_{hS} = \text{sample mean within post-stratum } h$ $s_{yh}^2 = \text{sample variance within post-stratum } h$ $N_h = \text{known population size of post-stratum } h$

One-Stage Cluster Sampling

$$s_t^2 = \frac{1}{n-1} \sum_{i \in S} (t_i - \bar{t}_S)^2 \quad s_r^2 = \frac{1}{n-1} \sum_{i \in S} (t_i - \hat{y}_r M_i)^2 = \frac{1}{n-1} \sum_{i \in S} M_i^2 (\bar{y}_i - \hat{y}_r)^2 \quad \bar{M}_S = \frac{1}{n} \sum_{i \in S} M_i$$

	Estimator	Estimated Variance	Confidence Interval
Total	$\hat{t} = N\bar{t}_S$	$\hat{V}(\hat{t}) = N^2 \frac{s_t^2}{n} (1 - \frac{n}{N})$	$\hat{t} \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\hat{t})}$
Mean	$\hat{\bar{y}} = \frac{\hat{t}}{M_0}$	$\hat{V}(\hat{\bar{y}}) = \frac{N^2 s_t^2}{n M_0^2} (1 - \frac{n}{N})$	$\hat{\bar{y}} \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\hat{\bar{y}})}$
Mean (ratio)	$\hat{\bar{y}}_r = \frac{\bar{t}_S}{\bar{M}_S}$	$\hat{V}(\hat{\bar{y}}_r) = (1 - \frac{n}{N}) \frac{s_r^2}{n \bar{M}_S^2}$	$\hat{\bar{y}}_r \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\hat{\bar{y}}_r)}$

N = number of clusters in population n = number of clusters sampled M_0 = total population elements (known) M_i = size of cluster i $\bar{M}_S$ = sample mean cluster size use $\hat{y}_r$ when M_0 unknown

Two-Stage Cluster Sampling

	Estimator
Total	$\hat{t} = \frac{N}{n} \sum_{i \in S} \hat{t}_i = \sum_{i \in S} \sum_{j \in S_i} \frac{N M_i}{n m_i} y_{ij}$
Mean	$\hat{\bar{y}} = \frac{\hat{t}}{M_0}$
Mean (ratio)	$\hat{\bar{y}}_r = \frac{\hat{t}}{\hat{M}_0} = \frac{\hat{t}}{N \bar{M}_S}$

m_i = number of elements sampled from cluster i M_i = total elements in cluster i $\hat{t}_i = \frac{M_i}{m_i} \sum_{j \in S_i} y_{ij}$ = estimated total for cluster i $\hat{M}_0 = N \bar{M}_S$ use $\hat{y}_r$ when M_0 unknown $\bar{M}_S = \frac{1}{n} \sum_{i \in S} M_i$

Unequal Probability Sampling (with Replacement)

	Estimator	Estimated Variance	Confidence Interval
Total	$\hat{t}_\psi = \frac{1}{n} \sum_{i \in \mathcal{W}} \frac{t_i}{\psi_i}$	$\hat{V}(\hat{t}_\psi) = \frac{1}{n(n-1)} \sum_{i \in \mathcal{W}} \left(\frac{t_i}{\psi_i} - \hat{t}_\psi \right)^2$	$\hat{t}_\psi \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\hat{t}_\psi)}$
Mean	$\hat{y}_\psi = \frac{\hat{t}_\psi}{M_0}$	$\hat{V}(\hat{y}_\psi) = \frac{\hat{V}(\hat{t}_\psi)}{M_0^2}$	$\hat{y}_\psi \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\hat{y}_\psi)}$
Mean (ratio)	$\hat{y}_\psi = \frac{\hat{t}_\psi}{\hat{M}_{0\psi}}$	$\hat{V}(\hat{y}_\psi) = \frac{1}{\hat{M}_{0\psi}^2} \cdot \frac{1}{n(n-1)} \sum_{i \in \mathcal{W}} \left(\frac{t_i}{\psi_i} - \hat{y}_\psi \frac{M_i}{\psi_i} \right)^2$	$\hat{y}_\psi \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\hat{y}_\psi)}$

$\psi_i = P(\text{unit } i \text{ selected on any single draw})$ $t_i = \text{total of } y \text{ within PSU } i$ $M_i = \text{size of PSU } i$ $M_0 = \text{known total population size}$ $\hat{M}_{0\psi} = \frac{1}{n} \sum_{i \in \mathcal{W}} \frac{M_i}{\psi_i}$ use ratio mean when M_0 unknown

Two-Phase Sampling (for Nonresponse)

SRS of n units drawn; n_R respond, n_M do not ($n_R + n_M = n$). A random subsample of νn_M nonrespondents is then contacted and assumed to all respond.

	Estimator
Mean	$\bar{y} = \frac{n_R}{n} \bar{y}_R + \frac{n_M}{n} \bar{y}_M$
Total	$\hat{t} = N\bar{y} = \frac{N}{n} \sum_{i \in \mathcal{S}_R} y_i + \frac{N}{n\nu} \sum_{i \in \mathcal{S}_M} y_i$

$\mathcal{S}_R = \text{set of respondents}$ $\mathcal{S}_M = \text{set of subsampled nonrespondents}$ $\bar{y}_R = \text{mean of respondents}$ $\bar{y}_M = \text{mean of subsampled nonrespondents}$

Adjusting Weights for Nonresponse

	Estimator
Total	$\hat{t}_\phi = \sum_{i \in \mathcal{R}} \tilde{\omega}_i y_i$
Mean	$\hat{y}_\phi = \frac{\hat{t}_\phi}{\sum_{i \in \mathcal{R}} \tilde{\omega}_i}$

$\mathcal{R} = \text{set of respondents}$ $\tilde{\omega}_i = 1/(\pi_i \phi_i)$ $\phi_i = P(\text{unit } i \text{ responds})$